POPULATION AND SYSTEMS ECOLOGY

Module map

- Module 15: Population and Systems Ecology
- Connected lab: lab-16_population-systems-ecology.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 15

Population and Systems Ecology

1. Population growth models
2. Limits and carrying capacity
3. Community interactions
4. Energy flow through trophic levels
5. Matter cycling and systems feedback

Lab evidence

lab-16_population-systems-ecology.md

POPULATION AND SYSTEMS ECOLOGY

Learning objectives

- Compare exponential and logistic population growth.
- Explain carrying capacity and limiting factors.
- Distinguish density-dependent and density-independent regulation.
- Use trophic levels and the 10 percent rule to explain energy limits.
- Contrast energy flow with nutrient cycling.

OBJECTIVE 1
Compare exponential and logistic population growth.

OBJECTIVE 2
Explain carrying capacity and limiting factors.

OBJECTIVE 3
Distinguish density-dependent and density-independent regulation.

OBJECTIVE 4
Use trophic levels and the 10 percent rule to explain energy limits.

OBJECTIVE 5
Contrast energy flow with nutrient cycling.

POPULATION AND SYSTEMS ECOLOGY

Topic sequence

- Population growth models: Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.
- Limits and carrying capacity: Exponential per-capita growth is constant; logistic per-capita growth slows near carrying capacity K. Total logistic growth peaks near K/2.
- Community interactions: Density-dependent effects vary with crowding. A freeze with equal proportional impact at low and high density is density-independent.
- Energy flow through trophic levels: Assume 10 percent transfer per trophic step for calculations; real efficiencies vary. From 10000 kJ in producers, primary consumers receive 1000 kJ.
- Matter cycling and systems feedback: Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

01

Population growth models

Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.

02

Limits and carrying capacity

Exponential per-capita growth is constant; logistic per-capita growth slows near carrying capacity K. Total logistic growth peaks near K/2.

03

Community interactions

Density-dependent effects vary with crowding. A freeze with equal proportional impact at low and high density is density-independent.

04

Energy flow through trophic levels

Assume 10 percent transfer per trophic step for calculations; real efficiencies vary. From 10000 kJ in producers, primary consumers receive 1000 kJ.

05

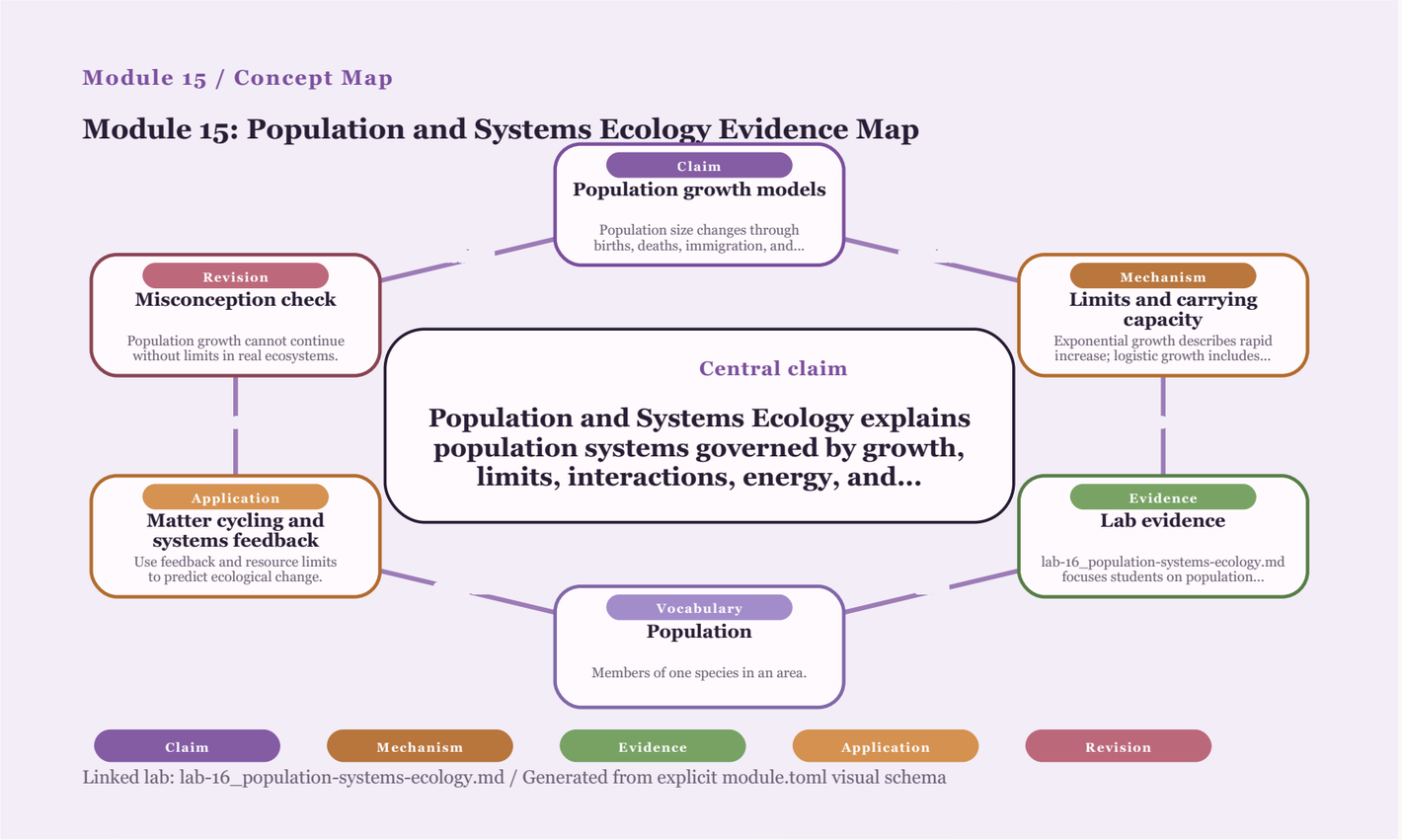
Matter cycling and systems feedback

Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

POPULATION AND SYSTEMS ECOLOGY

Module 15: Population and Systems Ecology Evidence Map

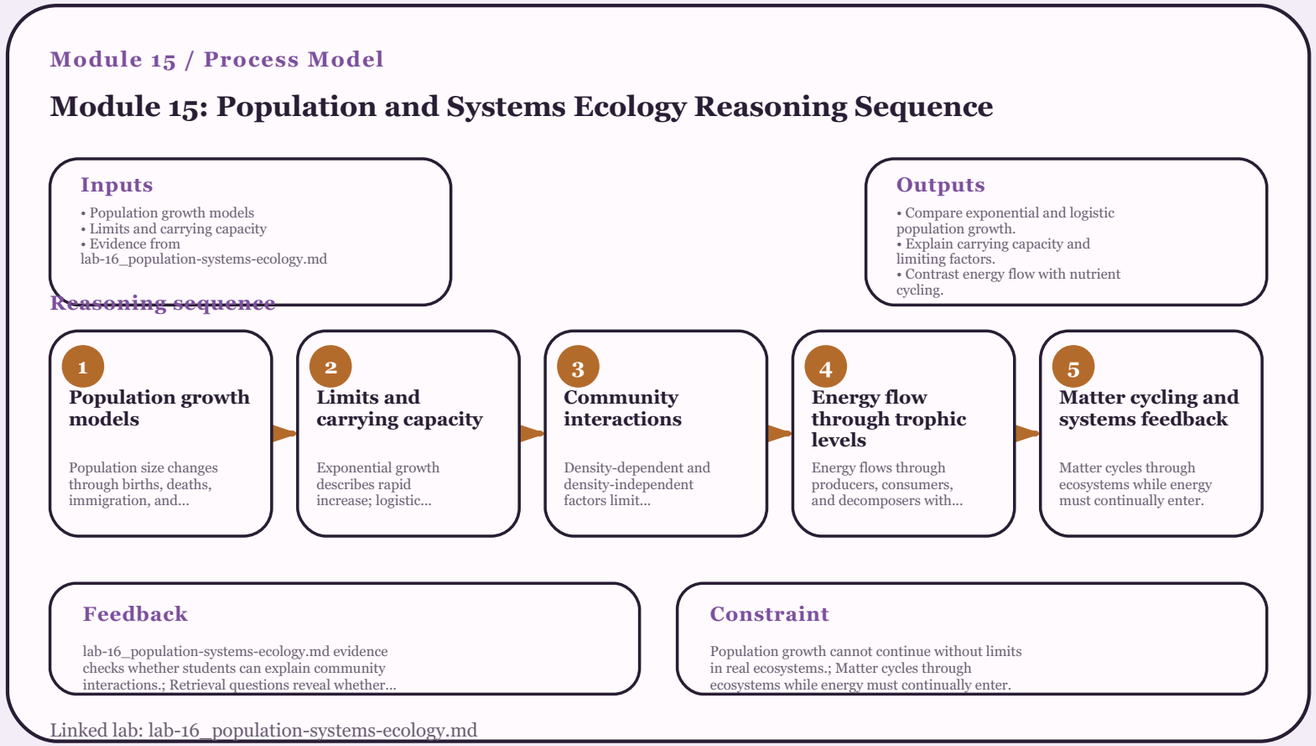
- Module focus: Population and Systems Ecology
- Central claim: Population and Systems Ecology explains population systems governed by growth, limits, interactions, energy, and feedback.
- Key nodes to connect: Population growth models, Limits and carrying capacity, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-16_population-systems-ecology.md supplies an evidence surface for the map.



POPULATION AND SYSTEMS ECOLOGY

Module 15: Population and Systems Ecology Reasoning Sequence

- Module focus: Population and Systems Ecology
- Trace the model: Population growth models -> Limits and carrying capacity -> Community interactions -> Energy flow through trophic levels
- Inputs become outputs: Population growth models, Limits and carrying capacity -> Compare exponential and logistic population growth., Explain carrying capacity and limiting factors.
- Feedback check: lab-16_population-systems-ecology.md evidence checks whether students can explain community interactions.
- Lab connection: lab-16_population-systems-ecology.md gives students a process to observe or test.



POPULATION AND SYSTEMS ECOLOGY

Terms as evidence handles

- Population: Members of one species in an area.
- Exponential growth: Growth pattern with increasing rate under abundant resources.
- Logistic growth: Growth that slows as population approaches carrying capacity.
- Carrying capacity: Approximate environment-supported population size.
- Density-dependent factor: Limiting factor whose impact changes with crowding.
- Density-independent factor: Limiting factor not directly tied to density.

POPULATION
Members of one species in an area.

EXPONENTIAL GROWTH
Growth pattern with increasing rate under abundant resources.

LOGISTIC GROWTH
Growth that slows as population approaches carrying capacity.

CARRYING CAPACITY
Approximate environment-supported population size.

DENSITY-DEPENDENT FACTOR
Limiting factor whose impact changes with crowding.

DENSITY-INDEPENDENT FACTOR
Limiting factor not directly tied to density.

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Lab connection

- Lab file: lab-16_population-systems-ecology.md
- Lab evidence should test or illustrate: Community interactions
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Population growth models

EVIDENCE
lab-16_population-systems-ecology.md

CLAIM
Compare exponential and logistic population growth.

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Contrast check

- Do not stop at: Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.
- Stronger explanation: Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Births and immigration add individuals; deaths and emigration remove them. Compare counts over the same area and time interval.

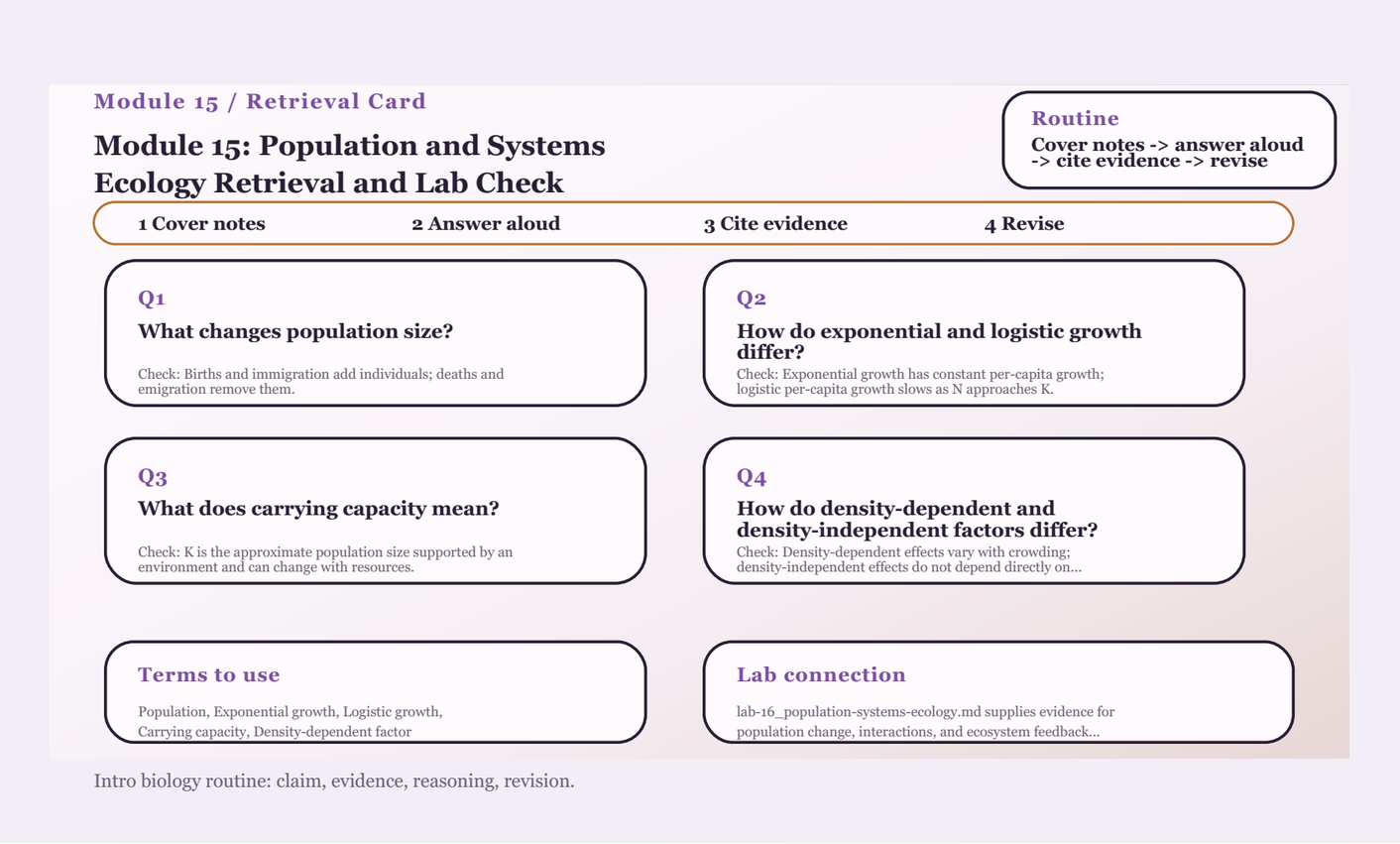
DEEPER

Matter cycles; usable energy must enter and dissipates as heat. Decomposers return nutrients. Competition and resource feedback connect populations.

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Module 15: Population and Systems Ecology Retrieval and Lab Check

- Module focus: Population and Systems Ecology
- Retrieval prompt: What changes population size?
- Answer check: Births and immigration add individuals; deaths and emigration remove them.
- Required terms: Population, Exponential growth, Logistic growth
- Lab connection: lab-16_population-systems-ecology.md; lab-16_population-systems-ecology.md supplies evidence for population change, interactions, and ecosystem feedback data.



Intro biology routine: claim, evidence, reasoning, revision.

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Practice quiz bridge

- 1. Logistic growth differs from exponential growth because logistic growth:
- 2. A contagious disease that spreads faster when a population is crowded is an example of a:
- 3. About how much of the energy at one trophic level typically passes to the next level up?
- 4. Energy and matter behave differently in an ecosystem because:

QUESTION 1

Answer first

Logistic growth differs from exponential growth because logistic growth:

QUESTION 2

Answer first

A contagious disease that spreads faster when a population is crowded is an example of a:

QUESTION 3

Answer first

About how much of the energy at one trophic level typically passes to the next level up?

QUESTION 4

Answer first

Energy and matter behave differently in an ecosystem because:

POPULATION AND SYSTEMS ECOLOGY

Synthesis and exit ticket

- Exit claim: explain population growth models using evidence from lab-16_population-systems-ecology.md.
- Must include: Population, Exponential growth, Logistic growth.
- Revision prompt: what evidence would change your explanation?

CLAIM
Population growth models

EVIDENCE
lab-16_population-systems-ecology.md

REVISION
What would change your mind?